

8. Write a program to implement stack using linked list.

```
#include <stdio.h>
#include <stdlib.h>

// Node structure
struct Node {
    int data;
    struct Node* next;
};

struct Node* top = NULL;

// Push operation
void push(int value) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));

    if(newNode == NULL) {
        printf("Stack Overflow\n");
        return;
    }

    newNode->data = value;
    newNode->next = top;
    top = newNode;

    printf("%d inserted into stack\n", value);
}

// Pop operation
void pop() {
    if(top == NULL) {
        printf("Stack Underflow\n");
        return;
    }

    struct Node* temp = top;
    printf("%d deleted from stack\n", top->data);
    top = top->next;
    free(temp);
}

// Peek operation
void peek() {
    if(top == NULL) {
        printf("Stack is Empty\n");
    } else {
```

```
    printf("Top element is %d\n", top->data);
}
}
```

// Display operation

```
void display() {
    struct Node* temp = top;

    if(temp == NULL) {
        printf("Stack is Empty\n");
        return;
    }

    printf("Stack elements are:\n");
    while(temp != NULL) {
        printf("%d\n", temp->data);
        temp = temp->next;
    }
}
```

// Main function (User Input)

```
int main() {
    int choice, value;

    while(1) {
        printf("\n--- STACK MENU ---\n");
        printf("1. Push\n");
        printf("2. Pop\n");
        printf("3. Peek\n");
        printf("4. Display\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice) {
            case 1:
                printf("Enter value to insert: ");
                scanf("%d", &value);
                push(value);
                break;

            case 2:
                pop();
                break;

            case 3:
                peek();
                break;
```

```
        case 4:
            display();
            break;

        case 5:
            exit(0);

        default:
            printf("Invalid choice!\n");
    }
}

return 0;
}
```